CSA6102 – DIGITAL FORENSICS AND CRIME INVESTIGATION

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EXP 1

1. Write a Python program to demonstrate Locard's Exchange Principle using file timestamps (metadata trace demo).

```python
# ============================================
# Locard's Exchange Principle - Metadata Trace Demo
# Google Colab Compatible
# ============================================

import os
import time
from datetime import datetime

filename = "digital_evidence.txt"

# --------------------------------------------
# Step 1: Create the evidence file
# --------------------------------------------
with open(filename, "w") as file:
    file.write("Original Digital Evidence\n")

print("Evidence file created.")

# Wait so timestamps will differ
time.sleep(3)

# --------------------------------------------
# Step 2: Capture original metadata
# --------------------------------------------
original = os.stat(filename)

print("\n========== ORIGINAL METADATA ==========")
print("File Name :", filename)
print("Last Access Time :", datetime.fromtimestamp(original.st_atime))
print("Last Modified Time :", datetime.fromtimestamp(original.st_mtime))
print("Metadata Size :", original.st_size, "bytes")

# --------------------------------------------
# Step 3: Simulate tampering
# --------------------------------------------
print("\nSimulating file modification...")

with open(filename, "a") as file:
    file.write("Unauthorized modification detected.\n")

# Wait a little
time.sleep(2)
```

```python
# ---------------------------------------------
# Step 4: Capture new metadata
# ---------------------------------------------
updated = os.stat(filename)

print("\n========== UPDATED METADATA ==========")
print("File Name :", filename)
print("Last Access Time :", datetime.fromtimestamp(updated.st_atime))
print("Last Modified Time :", datetime.fromtimestamp(updated.st_mtime))
print("Metadata Size :", updated.st_size, "bytes")

# ---------------------------------------------
# Step 5: Compare metadata
# ---------------------------------------------
print("\n========== METADATA COMPARISON ==========")

if original.st_mtime != updated.st_mtime:
    print("Modification Time Changed")
    print("Trace Detected: File has been modified.")
else:
    print("No modification detected.")

if original.st_size != updated.st_size:
    print("File Size Changed")
else:
    print("File Size Unchanged")

# ---------------------------------------------
# Step 6: Display file contents
# ---------------------------------------------
print("\n========== FILE CONTENT ==========")

with open(filename, "r") as file:
    print(file.read())

# ---------------------------------------------
# Step 7: Demonstrate Locard's Principle
# ---------------------------------------------
print("\n========== CONCLUSION ==========")
print("Locard's Exchange Principle states:")
print('"Every contact leaves a trace."')

print("\nIn this demonstration:")
print("- The file was accessed and modified.")
print("- The metadata (timestamps and size) changed.")
```

OUTPUT :

```
••• Evidence file created.

    ========== ORIGINAL METADATA ==========
    File Name : digital_evidence.txt
    Last Access Time : 2026-07-11 04:09:35.920771
    Last Modified Time : 2026-07-11 04:09:35.921771
    Metadata Size : 26 bytes

    Simulating file modification...

    ========== UPDATED METADATA ==========
    File Name : digital_evidence.txt
    Last Access Time : 2026-07-11 04:09:35.920771
    Last Modified Time : 2026-07-11 04:09:38.925992
    Metadata Size : 62 bytes

    ========== METADATA COMPARISON ==========
    Modification Time Changed
    Trace Detected: File has been modified.
    File Size Changed

    ========== FILE CONTENT ==========
    Original Digital Evidence
    Unauthorized modification detected.


    ========== CONCLUSION ==========
    Locard's Exchange Principle states:
    "Every contact leaves a trace."

    In this demonstration:
    - The file was accessed and modified.
    - The metadata (timestamps and size) changed.
    - These changes serve as digital traces that can be examined during forensic investigations.
```